

Xylofy AI Internship Week-2 Project

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Employee Attrition Prediction using Machine Learning Project Summary

1. Project Overview:

For my internship's second week's project at Xylofy AI, I worked on a machine learning classification problem to predict employee attrition. Employees changing jobs is just a normal part of business, but when a company unexpectedly loses its best talent, it causes a lot of stress, slows down projects, and costs a huge amount of money to fix. The goal of this project was to see if a machine learning algorithm could accurately predict whether an employee is likely to leave the company based on various workplace factors.

I used a dataset containing information on 1,470 employees. For each employee, the dataset provided the final attrition status (our target variable: Yes or No) along with 34 different features, such as monthly income, overtime status, age, job role, department, work-life balance rating, and years at the company.

2. My Step-by-Step Process:

To build a reliable prediction model, I followed a complete, end-to-end data science workflow:

Step 1: Data Loading and Inspection:

First, I imported the necessary Python libraries like Pandas, NumPy, Matplotlib, and Seaborn. I loaded the HR dataset into a Pandas DataFrame and looked at the first few rows to understand its structure. I used the `.info()` and `.describe()` functions to check the data types and get a statistical summary. I found a mix of numerical data (like age and income) and text-based categorical data (like job role and department). I also discovered that only 16% of employees left the company, meaning our target variable was highly imbalanced.

Step 2: Data Cleaning:

Before feeding data to an algorithm, it needs to be clean. I checked the dataset for missing values (nulls) and was fortunate to find none. I then dropped columns that provided no predictive value, such as EmployeeNumber (just a random ID), Over18 (which was 'Y' for everyone), and StandardHours (which was exactly 80 for everyone). I also converted our target Attrition column from 'Yes'/'No' into binary 1s and 0s.

Step 3: Exploratory Data Analysis (EDA):

This was one of the most interesting parts of the project because I got to visualize the data and look for patterns before doing any actual machine learning.

- **Department & Role Analysis:** I used bar charts to discover that the Sales Department, particularly Sales Representatives, had an alarming attrition rate of nearly 40%.
- **Income Analysis:** I used boxplots to compare monthly income against attrition, which showed a clear trend: employees who left generally earned noticeably less than those who stayed.
- **Tenure Analysis:** I plotted histograms showing a massive spike in employees quitting during their first 2 to 5 years at the company.

Step 4: Data Preprocessing:

Machine learning algorithms only understand numbers, not text. Since our dataset had categorical columns like "Department" and "JobRole," I used a technique called One-Hot Encoding (pd.get_dummies). I dropped the first column to avoid the dummy variable trap, preventing redundant data from confusing the model. I also used a StandardScaler to scale all the numerical features so large numbers (like income) wouldn't overpower smaller numbers (like years at the company).

Step 5: Handling Class Imbalance & Train-Test Split:

Because only 16% of the workforce left, a normal model might just guess "Stayed" every time and still be 84% accurate. To fix this, I used a technique called RandomOverSampler to artificially balance the data by randomly duplicating the minority class (the employees who left) until it was 50/50. After balancing the data, I split it into two parts: 80% was used for "training" the model, and 20% was kept hidden for "testing" to grade its true performance.

Step 6: Model Building and Evaluation:

I decided to test out three different classification models:

1. **Logistic Regression:** A basic, highly explainable model that serves as a strong baseline.
2. **Random Forest Classifier:** A complex, advanced algorithm that builds lots of decision trees and averages their guesses together.
3. **Gradient Boosting Classifier:** A powerful sequential tree-based algorithm.

I trained all three models on the 80% training data and tested them on the 20% testing data. To grade them, instead of just using Accuracy, I calculated Precision, Recall (how many real quitters the model actually caught), F1-Score, and ROC-AUC.

The evaluation results are:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.771	0.77	0.79	0.78	0.86
Gradient Boosting	0.87	0.88	0.91	0.89	0.94
Random Forest	0.962	0.96	0.98	0.97	0.99

3. Observations from the Data:

Based on exploring the data and evaluating the models, here are the main things I noticed about employee attrition:

- **Money is the Biggest Driver:** The Feature Importance chart revealed that a person's MonthlyIncome is the absolute strongest predictor of whether they will stay or quit.
- **The Burnout Factor (Overtime):** Right behind salary is OverTime. Employees who are forced to work extra hours on a regular basis are at a massive risk of quitting.

- **Age and Experience Matter:** Age and TotalWorkingYears are also top factors. Younger, less experienced employees tend to job-hop much more often than older, established professionals.
- **The 2-to-5 Year Danger Zone:** The time an employee has spent at the company (YearsAtCompany) is another strong trigger. If someone is going to leave, they usually do it during their first 2 to 5 years.
- **Complex Models succeed on Balanced Data:** Because we properly resampled the data using RandomOverSampler, complex models like Random Forest were able to fully capture the complex nuances of why employees leave, giving us near-perfect accuracy and recall.

4. Final Insights:

From this project, I learned exactly what makes an employee decide to leave a company. If an employee is not being paid enough (Monthly Income) and is simultaneously being overworked (Overtime), they are almost guaranteed to start looking for a new job.

If I were advising an HR Director based on this data, my main suggestions would be to review salaries specifically for employees in the 2-to-5 year tenure mark to ensure they are competitive. I would also strongly advise implementing an "Overtime Alert" system to prompt managers to balance workloads before top performers burn out.

Executive HR Summary (Non-Technical Overview):

Employee turnover is incredibly expensive because it disrupts teams and forces companies to spend a lot of time and money finding and training replacements. Through an analysis of 1,470 employee records, this project uncovered the hidden patterns behind why people decide to pack up and leave, and how HR departments can step in to retain their best talent before it's too late.

What's Actually Driving People Away?

With an overall attrition rate sitting at about 16%, the data clearly shows that the decision to quit usually comes down to employees weighing their paychecks against their stress levels. The top three drivers of turnover are:

1. Uncompetitive Compensation:

The absolute biggest red flag is monthly income. Employees who feel they are not being paid a market-competitive rate are the most likely to walk out the door.

2. The Overtime Trap:

Right behind the salary is excessive overtime. Employees who are consistently working beyond standard hours are significantly more likely to quit. High pay might keep them around for a bit, but prolonged burnout is a massive breaking point.

3. The 2-to-5 Year Danger Zone:

The company is losing a large chunk of its workforce during their first 2 to 5 years. If an employee can be kept engaged and fairly compensated past that 5-year mark, they tend to stick around for the long haul.

Where Should Focus Go First?

If a company wants to see an immediate bump in retention numbers, they need to look straight at the Sales Department. Specifically, Sales Representatives are churning at an alarming rate of nearly 40%. It is highly recommended that leadership sits down with that team to understand what their day-to-day actually looks like and find ways to relieve some of the pressure.

Strategic Recommendations for HR:

Based on what the data is telling us, there are two concrete changes that should be implemented:

1. Set up an "Overtime Alert" System:

HR should actively monitor for employees who are logging heavy overtime weeks consecutively. When an employee hits a certain threshold, it should automatically prompt their manager to have a "load-balancing" conversation to either redistribute their work or offer them some recovery time off.

2. Introduce an 18-Month Career Check-In:

Because that 2-to-5 year window is so risky, HR needs to catch people before they start feeling stagnant or underpaid. Introducing a mandatory career check-in around the 18-month anniversary is crucial. This shouldn't be a standard performance review. Instead, it should be a forward-looking conversation about their compensation, promotion path, and how they want to grow within the company.

Note:

While this predictive system does an excellent job of identifying who might be at risk of leaving based on historical trends, it's important to remember that data can't predict unpredictable life events like a spouse relocating or a sudden health issue. This machine learning tool is meant to act as an early-warning radar to help guide HR conversations, not replace human judgment and one-on-one check-ins.

5. Conclusion:

This project was a great way to experience a complete classification machine learning workflow from start to finish. I started by cleaning and visualizing the dataset to uncover the real reasons people quit, and then I prepared the data so the algorithms could read it.

By comparing Logistic Regression with complex ensemble models like Random Forest, I learned a crucial lesson about dealing with lopsided data. Because only 16% of the employees actually left, it was critical to balance the data first using Random Over Sampling. Once balanced, the Random Forest model emerged as the undisputed winner, providing incredible predictive accuracy and giving us a clear look (via Feature Importance) at exactly why it made those predictions.

Overall, this project showed me how helpful machine learning can be in the real corporate world. By using predictive models, HR teams can stop guessing and actually use data to step in and save their best employees before they even hand in their resignation letters

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